

99-CE
MATHS
Paper 2



HONG KONG EXAMINATIONS AUTHORITY

HONG KONG CERTIFICATE OF EDUCATION EXAMINATION 1999

MATHEMATICS PAPER 2

11.15 am – 12.45 pm (1½ hours)

Subject Code 180

- Label 1 Read carefully the instructions on the Answer Sheet and insert the information required (including the Subject Code) in the spaces provided.
- Label 2 When told to open this book, check that all the questions are there. Look for the words '**END OF PAPER**' after the last question.
- Label 3 **ANSWER ALL QUESTIONS.** All the answers should be marked on the Answer Sheet.
- Label 4 Note that you may only mark **ONE** answer to each question. Two or more answers will score **NO MARKS**.
- Label 5 All questions carry equal marks. No marks will be deducted for wrong answers.

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99-CE-MATHS 2-1

FORMULAS FOR REFERENCE

SPHERE	Surface area	$= 4\pi r^2$
	Volume	$= \frac{4}{3}\pi r^3$
CYLINDER	Area of curved surface	$= 2\pi rh$
	Volume	$= \pi r^2 h$
CONE	Area of curved surface	$= \pi rl$
	Volume	$= \frac{1}{3}\pi r^2 h$
PRISM	Volume	$= \text{base area} \times \text{height}$
PYRAMID	Volume	$= \frac{1}{3} \times \text{base area} \times \text{height}$

Figure 5



There are 36 questions in Section A and 18 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.

Section A

Label 1 If $f(x) = x^2 - 1$, then $f(a-1) =$

A. $a^2 - 2a$

B. $a^2 - 3a$

C. $a^2 - 3a - 2$

D. $a^2 = 1$

E. $a^2 - 2$

Label 2 $x^2 - y^2 - x + y =$

A. $(x - y)(x - y - 1)$

B. $(x - y)(x + y - 1)$

C. $(x - y)(x + y + 1)$

D. $(x + y)(x - y - 1)$

E. $(x + y)(x - y + 1)$

Label 3 If $a = \frac{1+b}{1-b}$, then $b =$

A! $\frac{a-1}{2}$

B! $\frac{a-1}{2a}$

C! $\frac{a+1}{a-1}$

D! $\frac{a-1}{a+1}$

E! $\frac{1-a}{a+1}$

Label 4 If $4^x = a$, then $16^x =$

A! $4a$

B! a^2

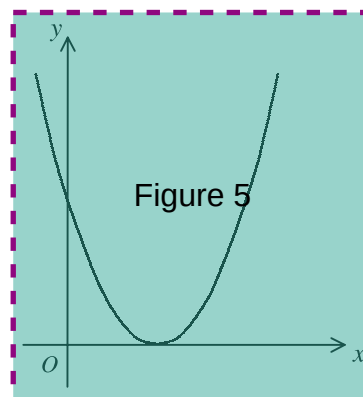
C! a^4

D! 2^a

E! 4^a

Label 5 In the figure, the graph of $y = x^2 - 6x + k$ touches the x -axis. Find k !

- A! $k \geq 0$
- B! $k \geq 9$
- C! $k = -9$
- D! $k = 0$
- E! $k = 9$



Label 6 If $(3x-1)(x-a) \equiv 3x^2 + bx - 2$, then

- A! $a \equiv 2, b \equiv -1$
- B! $a \equiv 2, b \equiv -7$
- C! $a \equiv -2, b \equiv 5$
- D! $a \equiv -2, b \equiv -5$
- E! $a \equiv -2, b \equiv -7$

Label 7 Solve $x^2 + 10x - 24 > 0$

A $x < -12$ or $x > 2$

B $x < -6$ or $x > -4$

C $x < -2$ or $x > 12$

D $-12 < x < 2$

E $-2 < x < 12$

Label 8 If $\begin{cases} y = x^2 + 3x - 2 \\ y = -x + 3 \end{cases}$ then

A $x = -1$

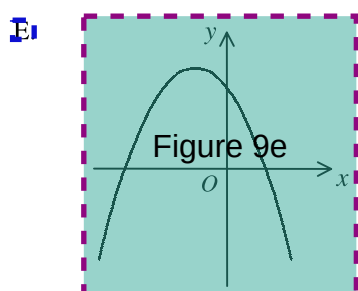
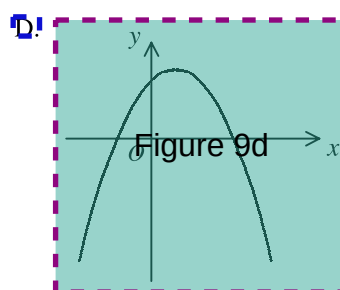
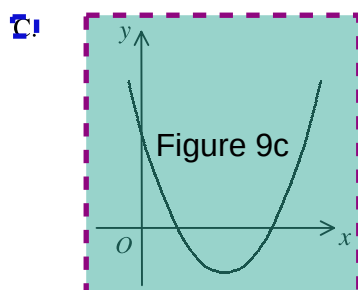
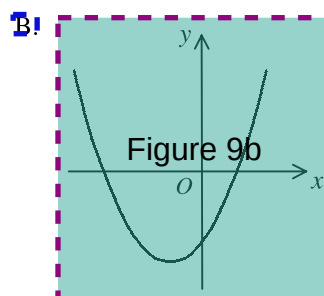
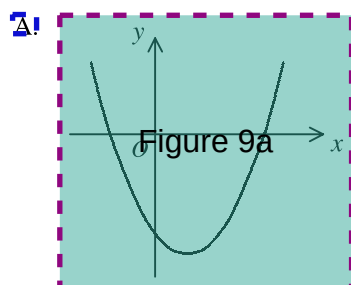
B $x = -1$ or 5

C $x = -2$ or 1

D $x = -3$ or 1

E $x = -3$ or 8

Label 9 Which of the following may represent the graph of $y = x^2 - 3x - 18$?



Label 10 The n -th term of an arithmetic sequence is $2 + 5n$. Find the sum of the first 100 terms of the sequence.

- A. 502
- B. 12450
- C. 25200
- D. 25450
- E. 25700

Label 11 In a class, students study either History or Geography, but not both. If the number of students studying Geography is 50% more than those studying History, what is the percentage of students studying History?

- A. 25%
- B. $33\frac{1}{3}\%$
- C. 40%
- D. 60%
- E. $66\frac{2}{3}\%$

Label 12. If $x:y = 3:4$ and $2x + 5y = 598$, find x .

- A. 23
- B. 26
- C. 69
- D. 78
- E. 104

Label 13. If 1 Australian dollar is equivalent to 4.69 H.K. dollars and 100 Japanese yen are equivalent to 5.35 H.K. dollars, how many Japanese yen are equivalent to 1 Australian dollar? Give your answer correct to the nearest Japanese yen.

- A. 4
- B. 25
- C. 88
- D. 114
- E. 2509

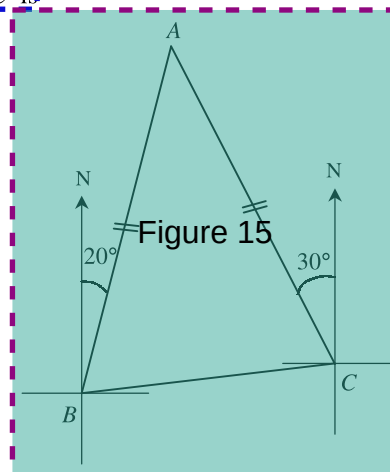
Label 14 Let m be a positive integer. Which of the following must be true?

- I. m^2 is even.
- II. $m(m+1)$ is even.
- III. $m(m+2)$ is even.

- A. I only
- B. II only
- C. III only
- D. I and III only
- E. II and III only

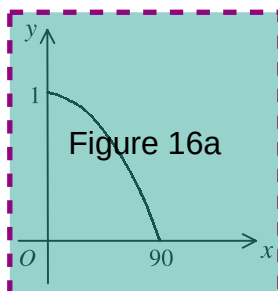
Label 15 In the figure, the bearing of B from C is

- A. $N5^\circ E$
- B. $N65^\circ E$
- C. $N85^\circ E$
- D. $S5^\circ W$
- E. $S85^\circ W$

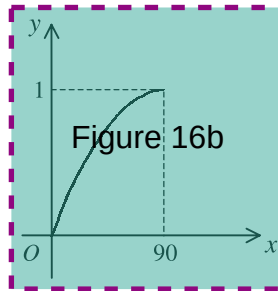


Label 16 Which of the following may represent the graph of $y = \cos x^\circ$ for $0 \leq x \leq 90$?

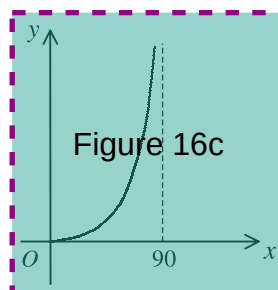
A



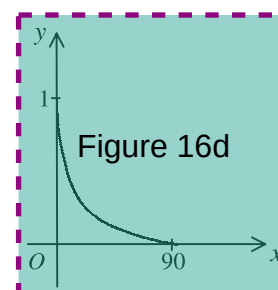
B



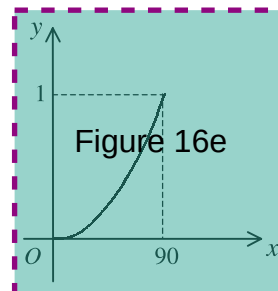
C



D

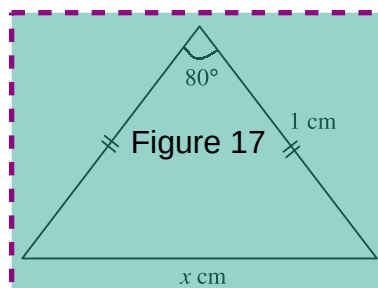


E



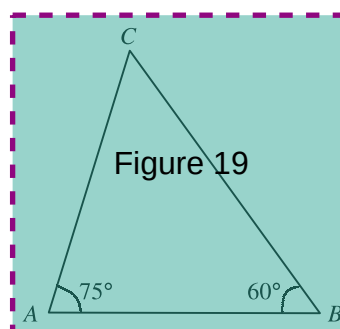
Label 17 In the figure, find x correct to 3 significant figures.

- A. 28
- B. 29
- C. 35
- D. 53
- E. 65



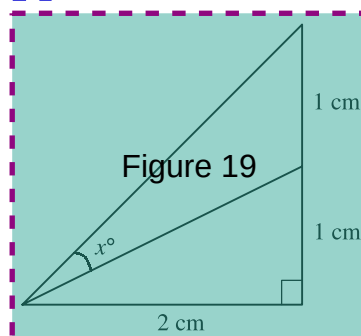
Label 18 Label 19 In the figure,

- A. $\frac{4}{3}$
- B. $\frac{5}{4}$
- C. $\frac{\sqrt{2}}{2}$
- D. $\frac{\sqrt{6}}{2}$
- E. $\frac{\sqrt{6}}{3}$



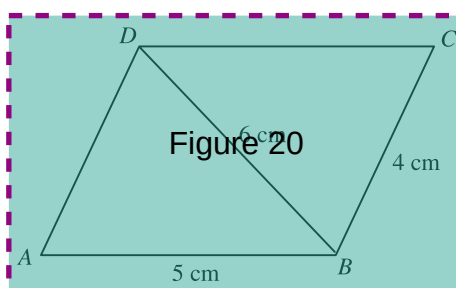
Label 19
19. In the figure, find x correct to 1 decimal place.

- A. 15.0
- B. 18.4
- C. 22.5
- D. 24.7
- E. 26.6



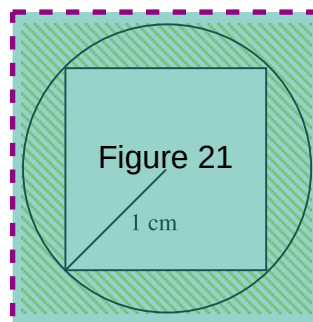
Label 20
20. In the figure, $ABCD$ is a parallelogram. Find $\angle ABC$ correct to the nearest degree.

- A. 83°
- B. 97°
- C. 104°
- D. 124°
- E. 139°



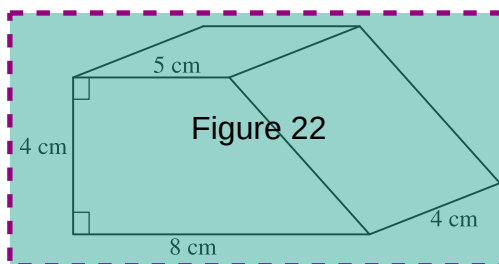
Label 21 In the figure, a square is inscribed in a circle with radius 1 cm . Find the area of the shaded region.

- A! $(\pi - 2) \text{ cm}^2$
- B! $(\pi - \sqrt{2}) \text{ cm}^2$
- C! $(\pi - 1) \text{ cm}^2$
- D! $(2\pi - 2) \text{ cm}^2$
- E! $(2\pi - 1) \text{ cm}^2$

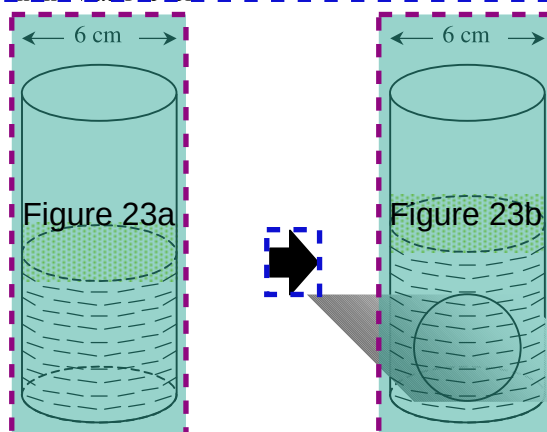


Label 22 The figure shows a right prism. Find its total surface area.

- A! 104 cm^2
- B! 108 cm^2
- C! 114 cm^2
- D! 120 cm^2
- E! 140 cm^2



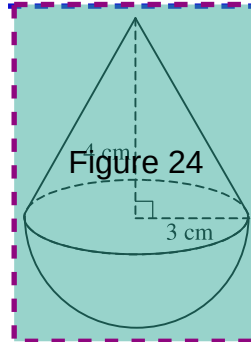
- Label 23
 In the figure, a cylindrical vessel of internal diameter 6 cm contains some water. A steel ball of radius 2 cm is completely submerged in the water. Find the rise in the water level.



- A. $\frac{32}{27}$ cm
 B. $\frac{8}{27}$ cm
 C. $\frac{16}{9}$ cm
 D. $\frac{4}{9}$ cm
 E. $\frac{8}{3}$ cm

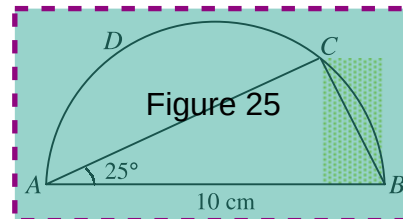
Label 24 In the figure, the solid consists of a right circular cone and a hemisphere with a common base. Find the volume of the solid.

- A! $30\pi \text{ cm}^3$
- B! $83\pi \text{ cm}^3$
- C! $48\pi \text{ cm}^3$
- D! $54\pi \text{ cm}^3$
- E! $72\pi \text{ cm}^3$



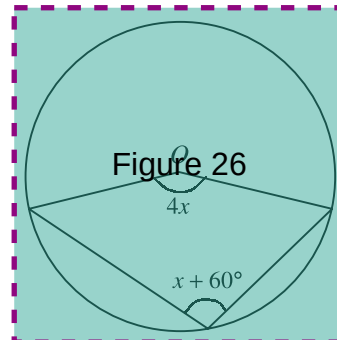
Label 25 In the figure, $ABCD$ is a semicircle. Find the area of the shaded region correct to the nearest 0.01 cm^2 .

- A! 5.33 cm^2
- B! 2.87 cm^2
- C! 2.67 cm^2
- D! 1.33 cm^2
- E! 0.17 cm^2



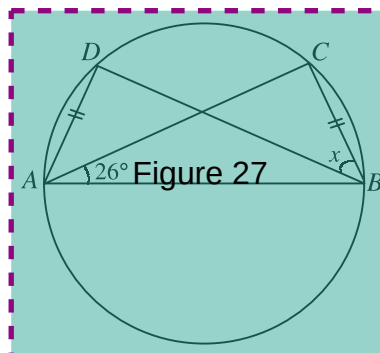
Label 26 In the figure, O is the centre of the circle. Find x .

- A! 12°
- B! 20°
- C! 24°
- D! 40°
- E! 60°



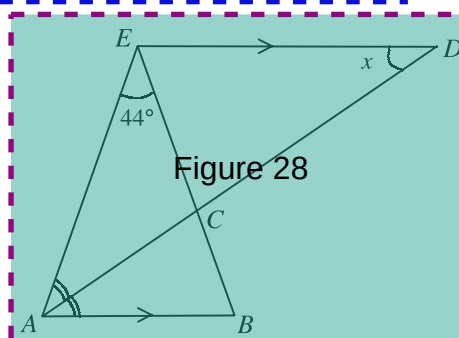
Label 27. In the figure, AB is a diameter of the circle. Find x .

- A. 26°
- B. 32°
- C. 38°
- D. 52°
- E. 64°



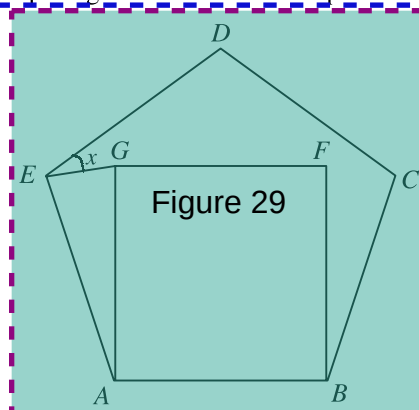
Label 28. In the figure, ACD and ECB are straight lines. If $\angle EAC = \angle CAB$ and $EA = EB$, find x .

- A. 22°
- B. 34°
- C. 44°
- D. 46°
- E. 68°



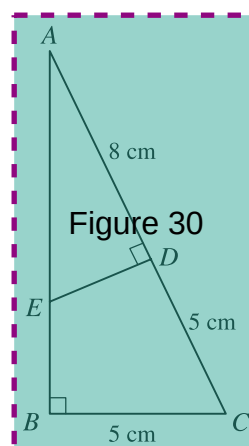
Label 29 In the figure, $ABCDE$ is a regular pentagon and $ABFG$ is a square. Find x .

- A. 18°
- B. 27°
- C. 30°
- D. 36°
- E. 45°



Label 30 In the figure, AEB and ADC are straight lines. Find ED .

- A. $\frac{10}{3}$ cm
- B. $\frac{40}{13}$ cm
- C. 3 cm
- D. $\sqrt{40}$ cm
- E. $\sqrt{80}$ cm



Label 31: $A(-4, 2)$ and $B(1, -3)$ are two points. C is a point on the y -axis such that $AC = CB$. Find the coordinates of C .

A: $(-\frac{3}{2}, -\frac{1}{2})$

B: $(-1, 0)$

C: $(1, 0)$

D: $(0, -1)$

E: $(0, 1)$

Label 32: In the figure, $OABC$ is a parallelogram. If the equation of OC is $2x - y = 0$ and the length of CB is 3, find the equation of AB .

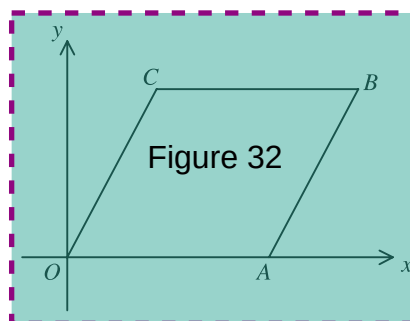
A: $x - 2y - 3 = 0$

B: $2x - y - 3 = 0$

C: $2x - y + 3 = 0$

D: $2x - y - 6 = 0$

E: $2x - y + 6 = 0$



Label 33 Find the median and mode of the ten numbers
6, 8, 3, 3, 5, 5, 5, 7, 7, 11.

- A! median = 5, mode = 5
- B! median = 5, mode = 5.5
- C! median = 5.5, mode = 5
- D! median = 5.5, mode = 6
- E! median = 6, mode = 5

Label 34 A student scored 50 marks in a test and the corresponding standard score is -0.5 . If the mean of the test scores is 60 marks, find the standard deviation of the scores.

- A! $\sqrt{20}$ marks
- B! 5 marks
- C! 9.5 marks
- D! 10 marks
- E! 20 marks

Label 35

Two cards are drawn randomly from four cards numbered 1, 2, 3 and 4 respectively. Find the probability that the sum of the numbers drawn is odd.

A. $\frac{1}{6}$

B. $\frac{1}{4}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

E. $\frac{2}{3}$

Label 36

Tom and Mary each throws a dart. The probability of Tom's dart hitting the target is $\frac{1}{3}$ while that of Mary's is $\frac{2}{5}$. Find the probability of only one dart hitting the target.

A. $\frac{2}{15}$

B. $\frac{3}{15}$

C. $\frac{7}{15}$

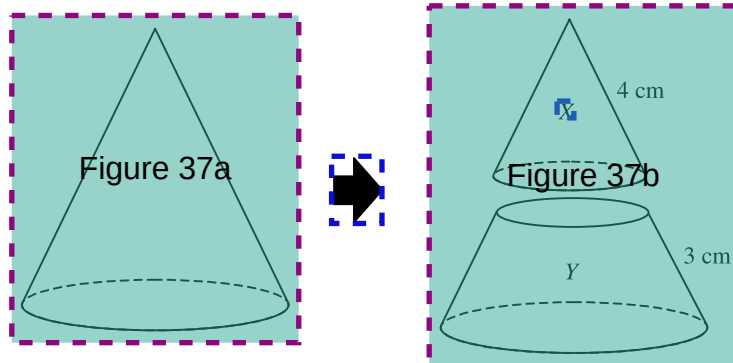
D. $\frac{11}{15}$

E. $\frac{13}{15}$



Section B

- Label 37 In the figure, a right circular cone is divided into two parts X and Y by a plane parallel to the base such that the lengths of their slant edges are 4 cm and 3 cm respectively. Find the ratio of the curved surface areas of X and Y .



- Label 38 It is given that $F(x) = x^3 - 4x^2 + ax + b$. $F(x)$ is divisible by $x - 1$. When it is divided by $x + 1$, the remainder is 12. Find a and b .

- A. $a = 5, b = 10$
 B. $a = 1, b = 2$
 C. $a = -3, b = 6$
 D. $a = -4, b = 7$
 E. $a = -7, b = 10$

Label 39 If $\frac{1}{2} \log y = 1 + \log x$, then

- A. $y = \sqrt{10x}$
- B. $y = 100 + x^2$
- C. $y = (10 + x)^2$
- D. $y = 10x^2$
- E. $y = 100x^2$

Label 40 $\frac{2}{x^2 - 1} - \frac{x - 1}{x^2 - 2x - 3} =$

- A. $\frac{-x^2 + 2x + 5}{(x - 1)(x + 1)(x + 3)}$
- B. $\frac{-x^2 + 2x + 7}{(x - 1)(x + 1)(x + 3)}$
- C. $\frac{-x^2 - 5}{(x - 3)(x - 1)(x + 1)}$
- D. $\frac{x^2 - 5}{(x - 3)(x - 1)(x + 1)}$
- E. $\frac{-x^2 + 4x - 7}{(x - 3)(x - 1)(x + 1)}$

41. The method of bisection is used to find the root of $\sin x + x - 1 = 0$ starting with the interval $[0, 2]$. After the first approximation, the interval which contains the root becomes $[0, 1]$. Find the interval which contains the root after the third approximation.

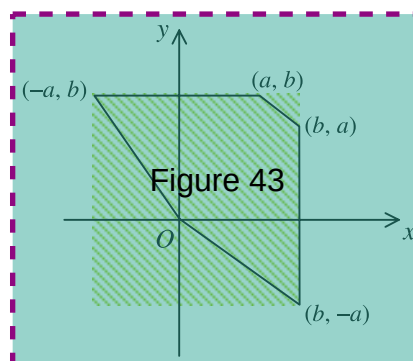
- A. $[0, 0.25]$
- B. $[0.25, 0.75]$
- C. $[0.5, 0.75]$
- D. $[0.5, 1]$
- E. $[0.75, 1]$

Label 42. John goes to school and returns home at speeds x km/h and $(x + 1)$ km/h respectively. The school is 2 km from John's home and the total time for the two journeys is 54 minutes. Which of the following equations can be used to find x ?

- A. $\frac{x}{2} + \frac{x+1}{2} = \frac{54}{60}$
- B. $\frac{2}{x} + \frac{2}{x+1} = \frac{54}{60}$
- C. $\frac{\frac{1}{2}(x+1)}{4} = \frac{54}{60}$
Figure 42a
- D. $\frac{\frac{4}{1}[x + (x+1)]}{2} = \frac{54}{60}$
Figure 42b
- E. $2x + 2(x+1) = \frac{54}{60}$

Label 43 In the figure, find the point (x, y) in the shaded region (including the boundary) at which $bx - ay + 3$ attains its greatest value.

- A! $(0, 0)$
- B! $(-a, b)$
- C! (a, b)
- D! $(b, -a)$
- E! (b, a)



Label 44 The sum of the first two terms of a geometric sequence is 3 and the sum to infinity of the sequence is 4. Find the common ratio of the sequence.

- A! $-\frac{1}{7}$
- B! $\frac{1}{7}$
- C! $\frac{1}{4}$
- D! $-\frac{1}{2}$
- E! $-\frac{1}{2}$ or $\frac{1}{2}$

Label 45 It is given that y varies inversely as x^3 . If x is increased by 100% then y is

- A. increased by 800%
- B. increased by 700%
- C. decreased by 300%
- D. decreased by 87.5%
- E. decreased by 12.5%

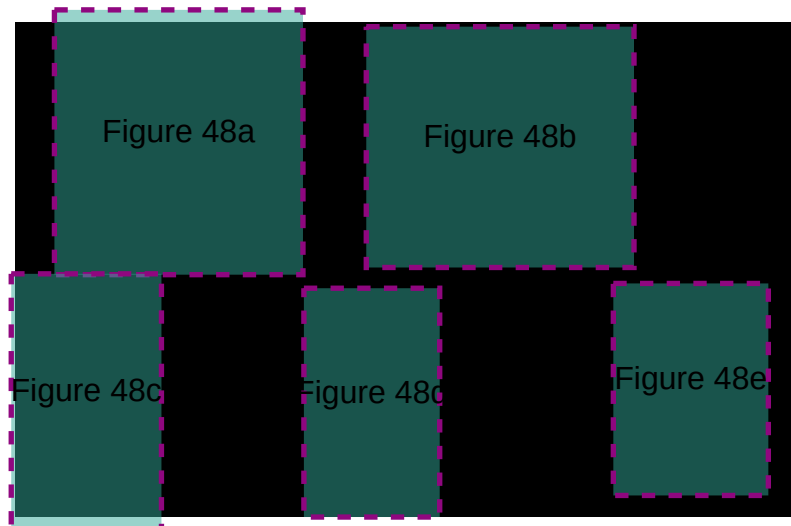
Label 46 $\frac{\cos(90^\circ - A) \cos(-A)}{\sin(360^\circ - A)} =$

- A. $-\cos A$
- B. $\cos A$
- C. $\sin A$
- D. $-\frac{\cos^2 A}{\sin A}$
- E. $\frac{\cos^2 A}{\sin A}$

Label 47 If $0 \leq \theta \leq 2\pi$, solve $(\cos \theta - 3)(3 \sin \theta - 2) = 0$ correct to 3 significant figures.

- A. 0.730 or 1.23
- B. 0.730 or 2.41
- C. 0.730 or 3.87
- D. 0.730 or 6.21
- E. 0.734 or 2.41

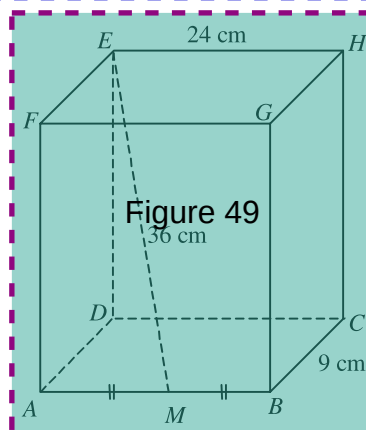
Label 48 The figure shows five sectors. Which of the marked angles measures 2π radians?



- A. 2π
- B. $\frac{\pi}{2}$
- C. $\frac{\pi}{4}$
- D. $\frac{\pi}{2}$
- E. $\frac{\pi}{4}$

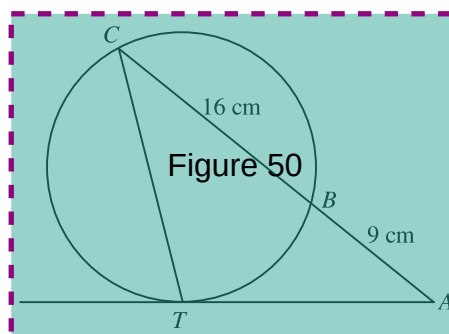
Label 49 In the figure, $ABCDEFGH$ is a rectangular block. Find the inclination of EM to the plane $ABCD$ correct to the nearest degree.

- A. 23°
- B. 25°
- C. 65°
- D. 71°
- E. 75°



Label 50 In the figure, AT is tangent to the circle at T and ABC is a straight line. Find AT .

- A. 9 cm
- B. 12 cm
- C. 15 cm
- D. 16 cm
- E. 20 cm



Label 51. In the figure, find the equation of the circle with AB as a diameter.

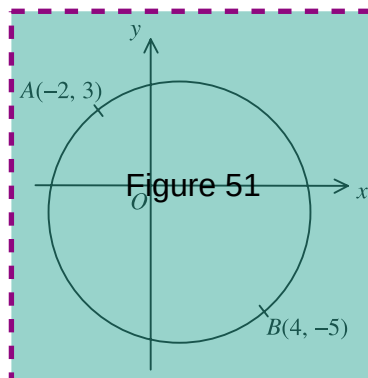
A. $x^2 + y^2 - 2x + 2y - 23 = 0$

B. $x^2 + y^2 - 2x + 2y - 3 = 0$

C. $x^2 + y^2 + 2x - 2y - 23 = 0$

D. $x^2 + y^2 + 2x - 2y - 3 = 0$

E. $x^2 + y^2 - 25 = 0$



Label 52. The figure shows a circle centred at C and passing through $O(0, 0)$, $A(6, 0)$ and $B(0, 8)$. Which of the following must be true?

I. C lies on the line $\frac{x}{6} + \frac{y}{8} = 1$.

II. The radius of the circle is 10.

III. OC is perpendicular to AB .

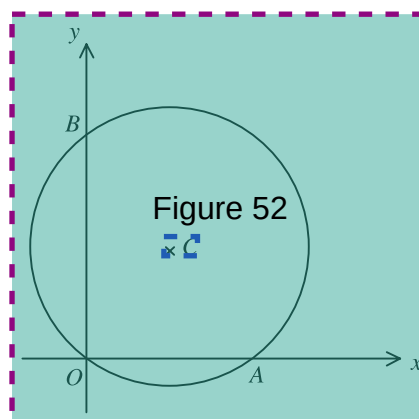
A. I only

B. II only

C. I and II only

D. I and III only

E. I, II and III



Label 53 Two circles with equations $(x+1)^2 + (y+1)^2 = 25$ and $(x-11)^2 + (y-8)^2 = 100$ touch each other externally at a point P . Find the coordinates of P .

A. $(-3, -2)$

B. $(\frac{7}{5}, \frac{4}{5})$

C. $(3, 2)$

D. $(5, \frac{7}{2})$

E. $(7, 5)$

Label 54 In the figure, $ABCD$ is a rectangle. M is the midpoint of BC and AC intersects MD at N .
Area of $\triangle NCD$: area of $\triangle BMN =$

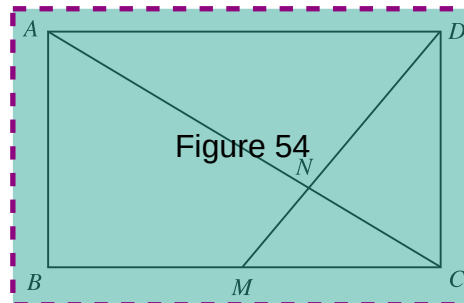
A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. $\frac{2}{5}$

E. $\frac{4}{7}$



END OF PAPER

1999 Mathematics (Paper 2)

Question No.	Key	Question No.	Key
1.	A	31.	E
2.	B	32.	D
3.	D	33.	C
4.	B	34.	E
5.	E	35.	E
6.	C	36.	C
7.	A	37.	B
8.	D	38.	E
9.	A	39.	E
10.	D	40.	E
11.	C	41.	C
12.	C	42.	B
13.	C	43.	D
14.	B	44.	E
15.	E	45.	D
16.	A	46.	A
17.	B	47.	B
18.	D	48.	E
19.	B	49.	C
20.	B	50.	C
21.	A	51.	A
22.	E	52.	A
23.	A	53.	C
24.	A	54.	D
25.	D		
26.	D		
27.	C		
28.	B		
29.	B		
30.	A		